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<https://dasma.pcu.edu.ph>



Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SHH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo system ctl start ssh` – starting the SHH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:          0.0
Usage of /:           5.1% of 48.91GB
Memory usage:         5%
Swap usage:           0%
Processes:            139
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- Server IP address (output of `ip a` or `hostname -I`)
- SSH service status (`sudo systemctl status ssh`)
- Successful SSH connection from host
- Verification commands executed inside VM

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Student ID: 202201528

Course/Year/Section: BSIT3B

Date Submitted: 9/2/26

Instructor: Johneros P. Puyo

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☐ **Create and configure a Virtual Machine with Ubuntu Server**
- ☐ **Configure Bridged Network Adapter for network access**
- ☐ **Determine and verify the server's IP address**
- ☐ **Install OpenSSH Server and enable the service**
- ☐ **Establish successful SSH connection from host machine**
- ☐ **Execute verification commands to confirm remote access**

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Ubuntu
Virtualization Software (VirtualBox/VMware)	Oracle VM VirtualBox 7.0
Ubuntu Server Version	Ubuntu server 26.04LTS
VM Name	sarcaogaS
RAM Allocated	5612MB
Storage Allocated	100 GB
Network Mode	Host-only Adapter
SSH Client Used	Secure shell client

3 VM CONFIGURATION

Setting	Value
VM Name	UBUNTU-SERVER
OS Type	Ubuntu (64-bit)
Memory	5612 MB
Hard Disk	100 GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

Commands Executed:

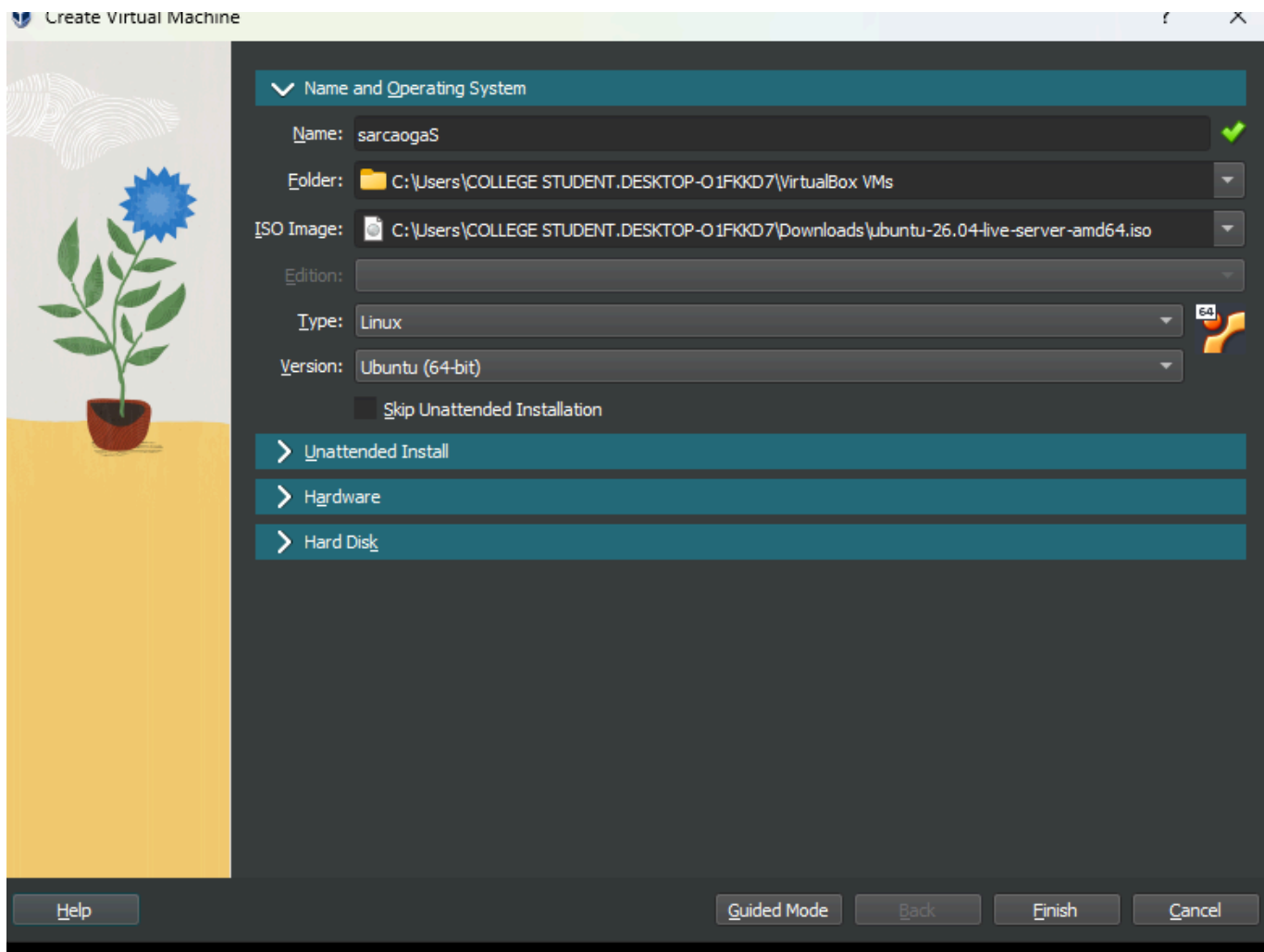
Server IP Address: _____

Network Interface: _____

6. REMOTE CONNECTION

Connection Details:

Parameter	Value
Server IP	192.168.137.89
Username	sarcaoga
SSH Port	ssh sarcaoga@192.168.137.89



Create Virtual Machine

> Name and Operating System

< Unattended Install

Username and Password

Additional Options

Username: vboxuser ✓

Password: ●●●●●● ●

Repeat Password: ●●●●●● ●

Product Key: #####-#####-#####-#####-###

Hostname: sarcaogaS ✓

Domain Name: myguest.virtualbox.org

☐ Install in Background

Guest Additions

Guest Additions ISO: <not selected>

> Hardware

> Hard Disk

Help

Guided Mode

Back

Finish

Cancel



Create Virtual Machine



> Name and Operating System

> Unattended Install

< Hardware

Base Memory:

4 MB

16384 MB

 5685 MB

Processors:

1 CPU

12 CPUs

 4

☐ Enable EFI (special OSes only)

> Hard Disk

Help

Guided Mode

Back

Finish

Cancel

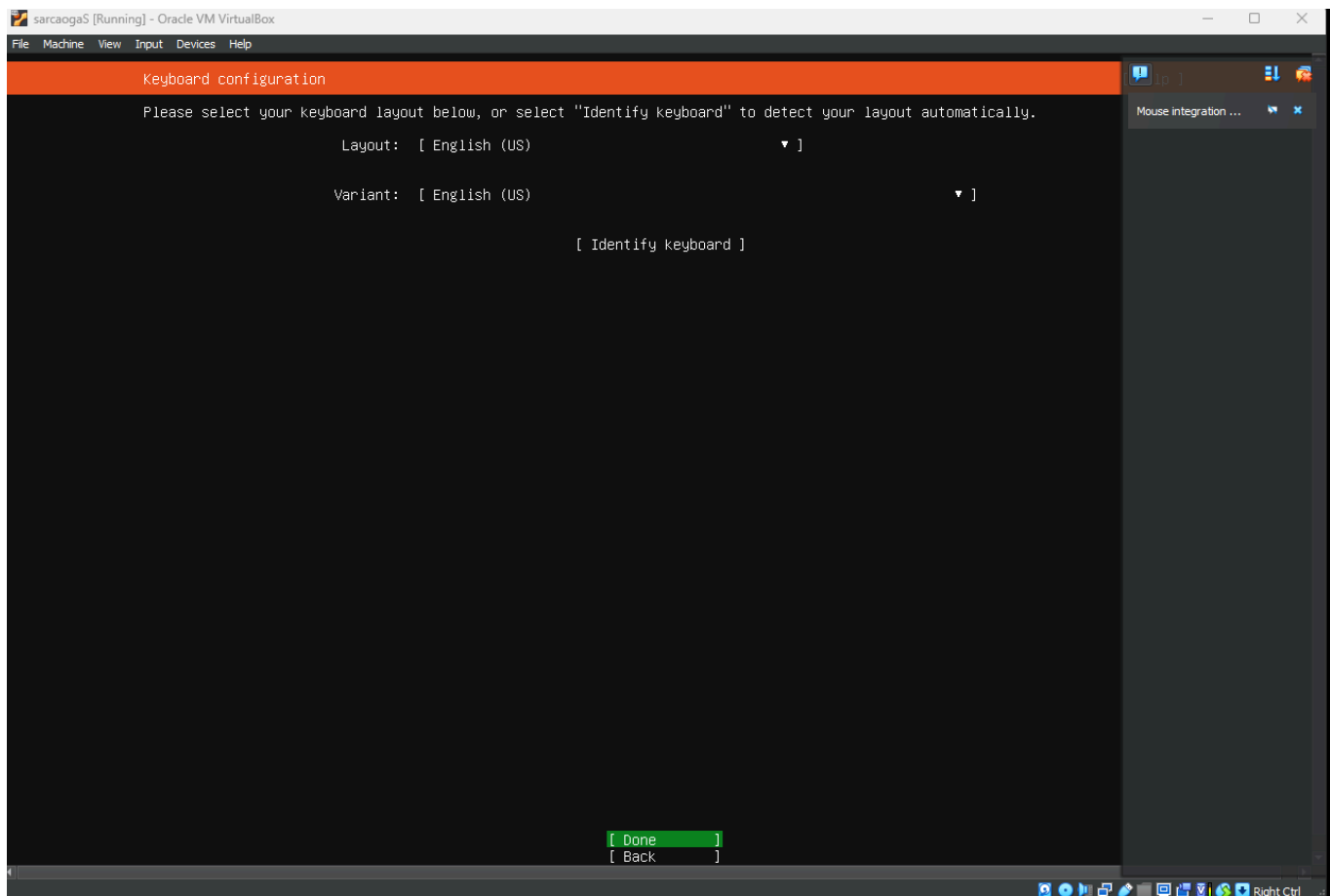
Willkommen! Bienvenue! Welcome! Добро пожаловать! Welkom!

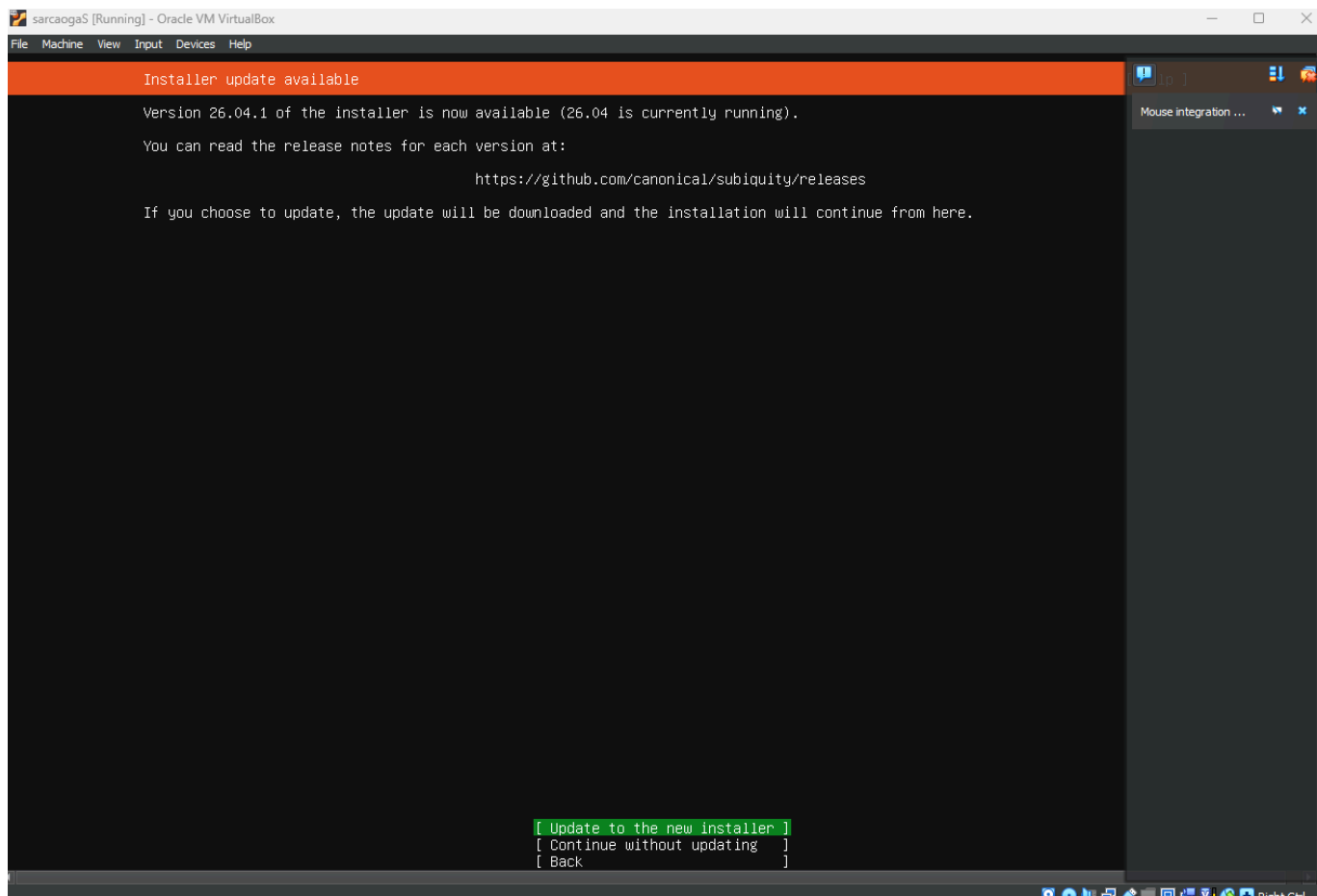
Use UP, DOWN and ENTER keys to select your language.

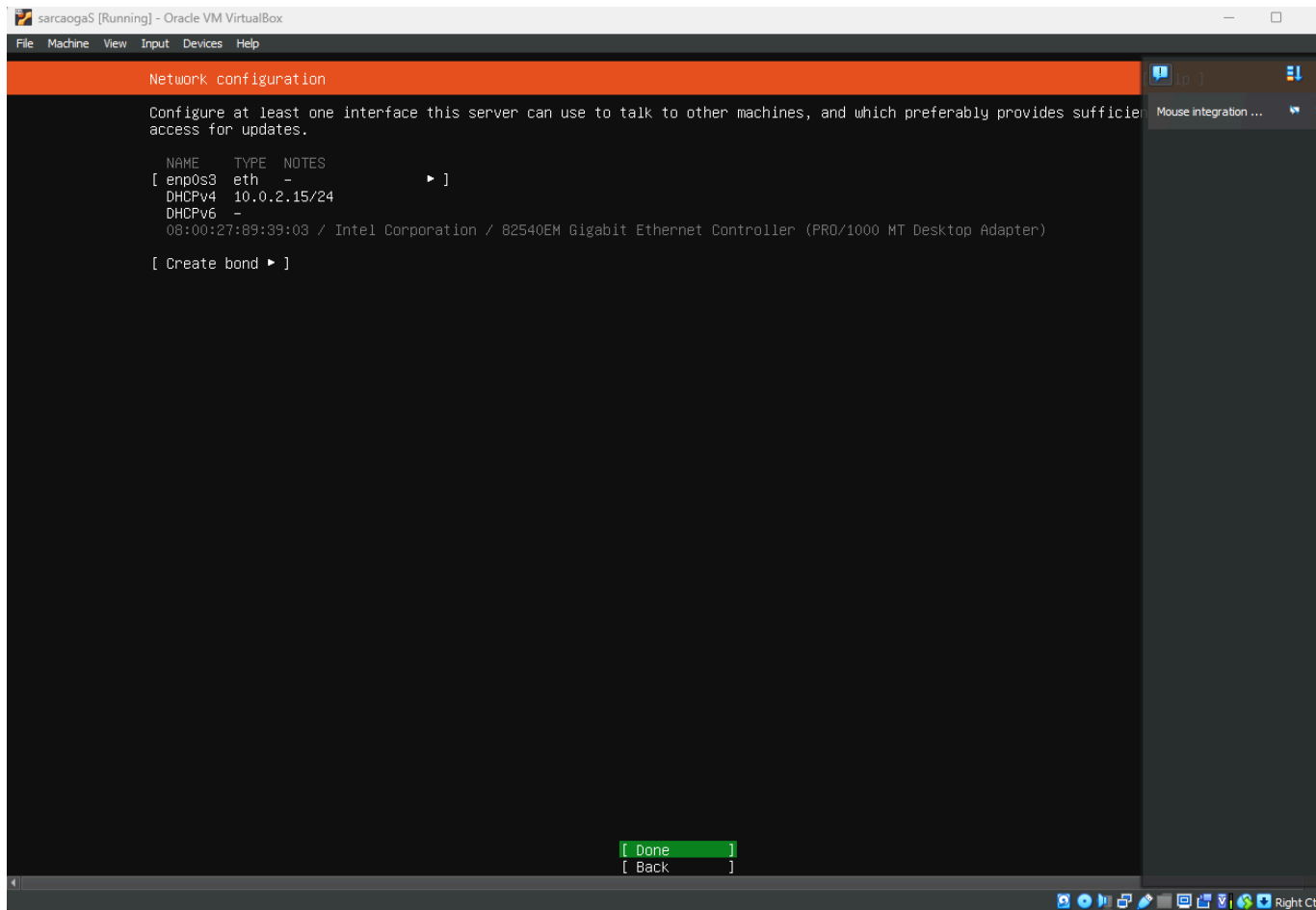
```
[ Asturianu
[ Bahasa Indonesia
[ Català
[ Deutsch
[ English
[ English (UK)
[ Español
[ Français
[ Galego
[ Hrvatski
[ Latviski
[ Lietuviškai
[ Magyar
[ Nederlands
[ Norsk bokmål
[ Occitan
[ Polski
[ Português
[ Suomi
[ Svenska
[ Čeština
[ Ελληνικά
[ Беларуская
[ Русский
[ Српски
[ Українська
```

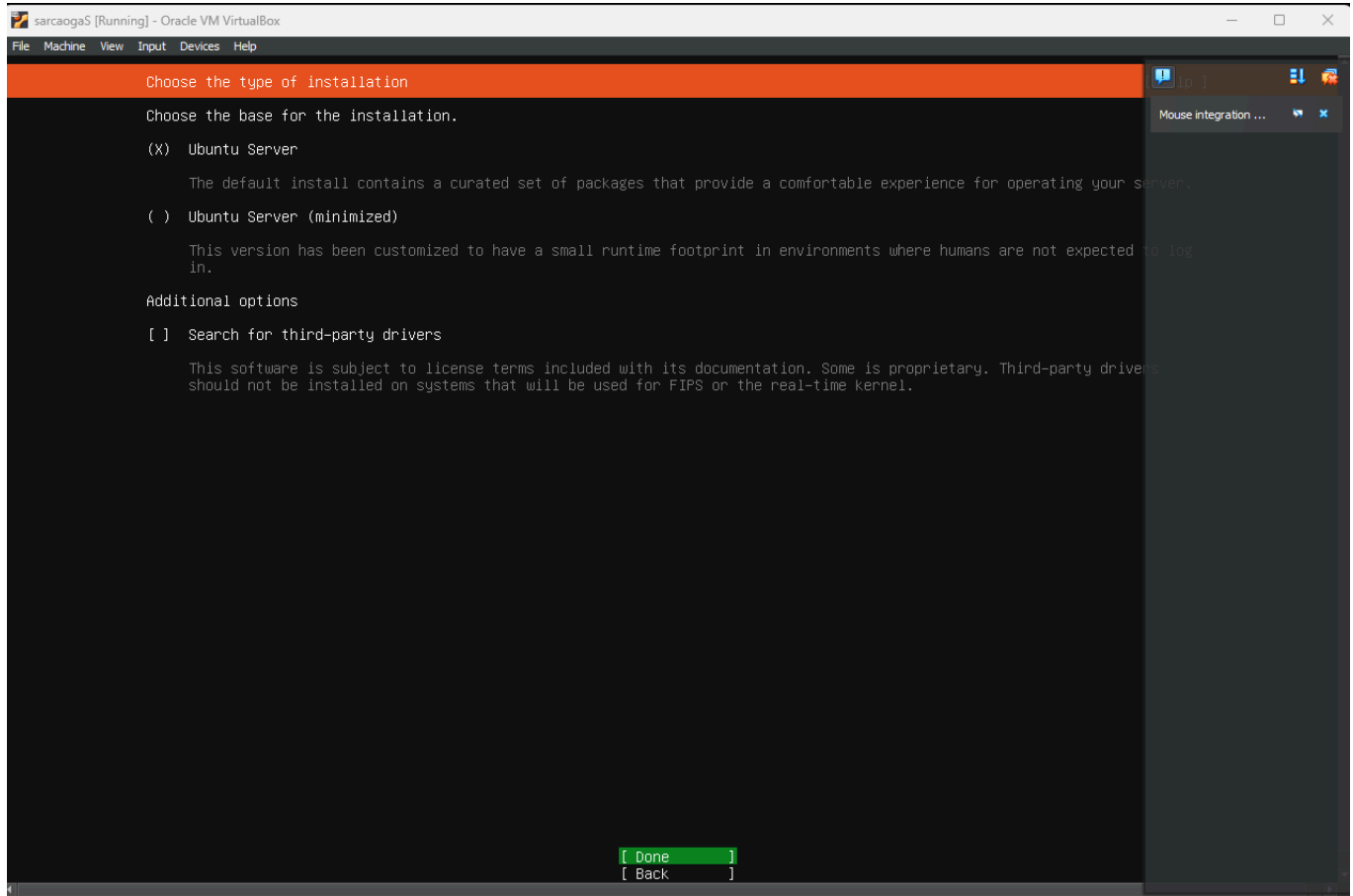
ip }

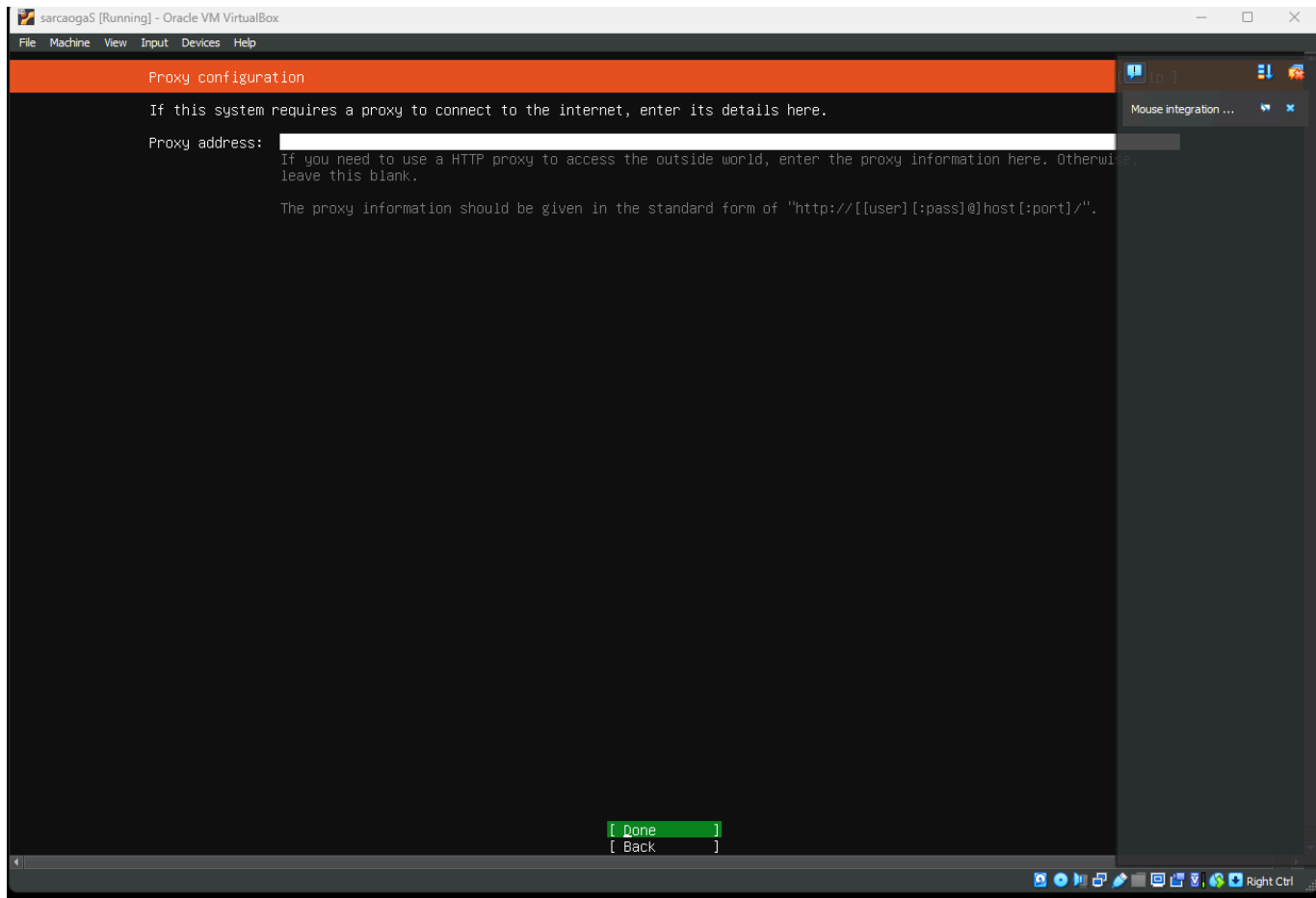
Mouse integration ...

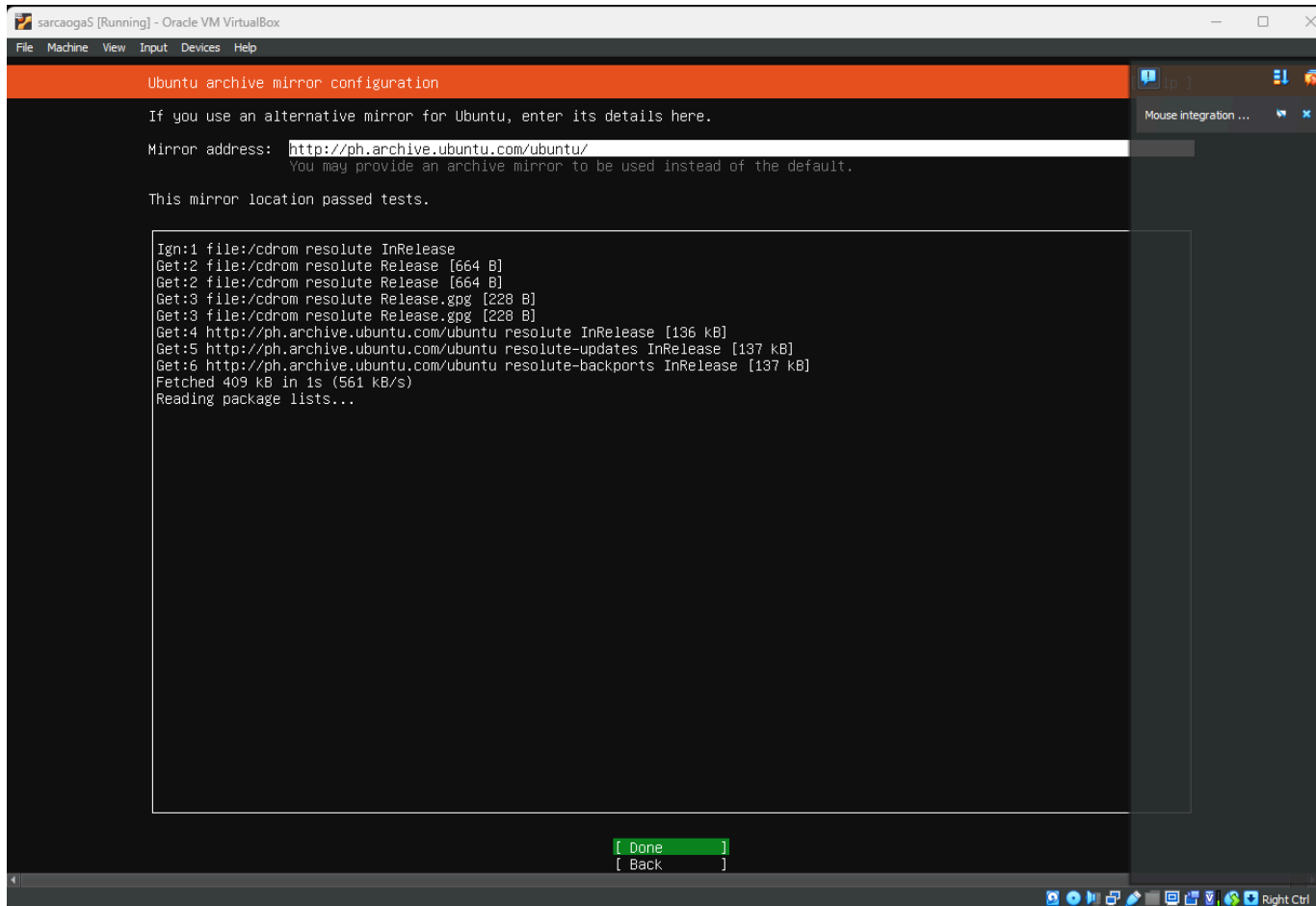


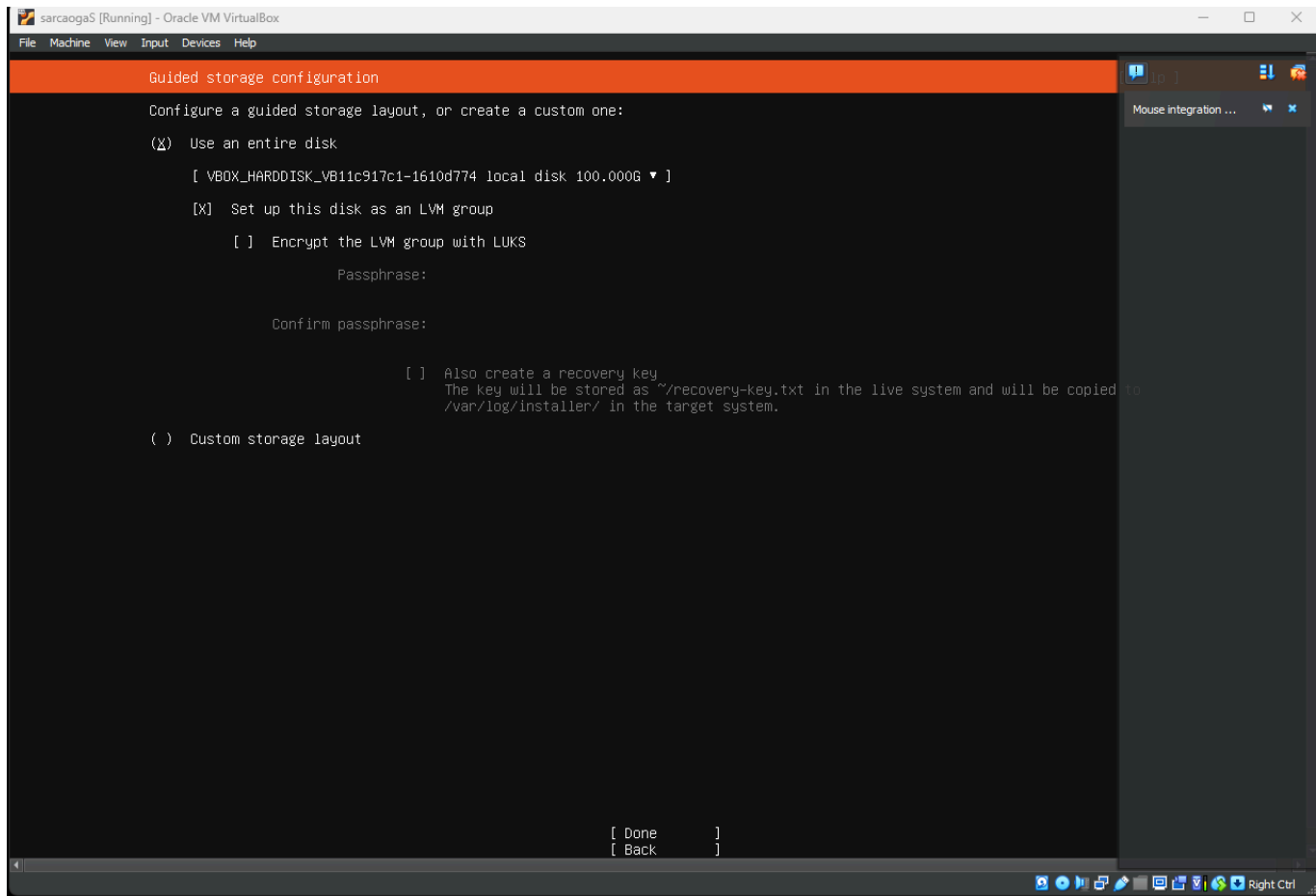


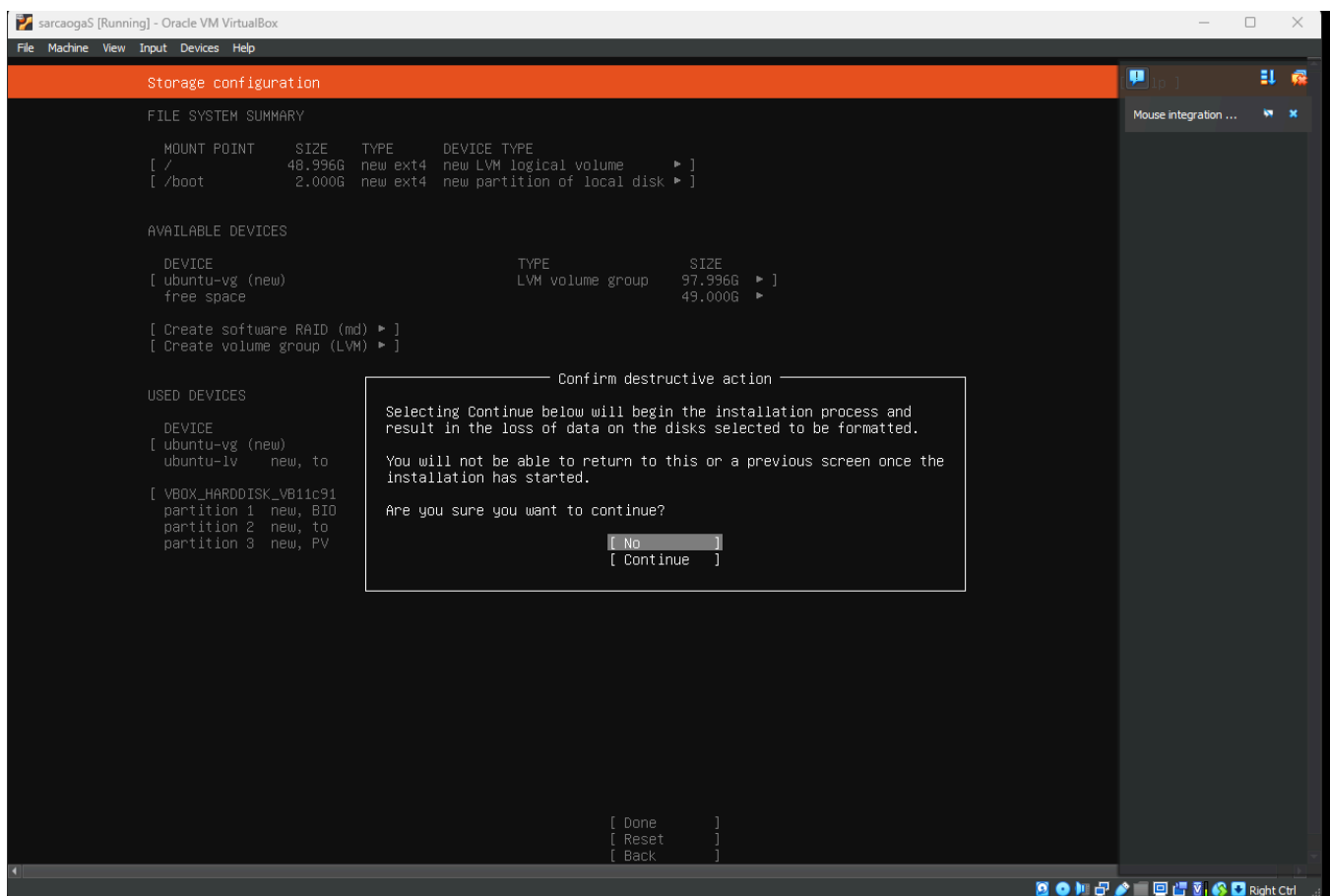
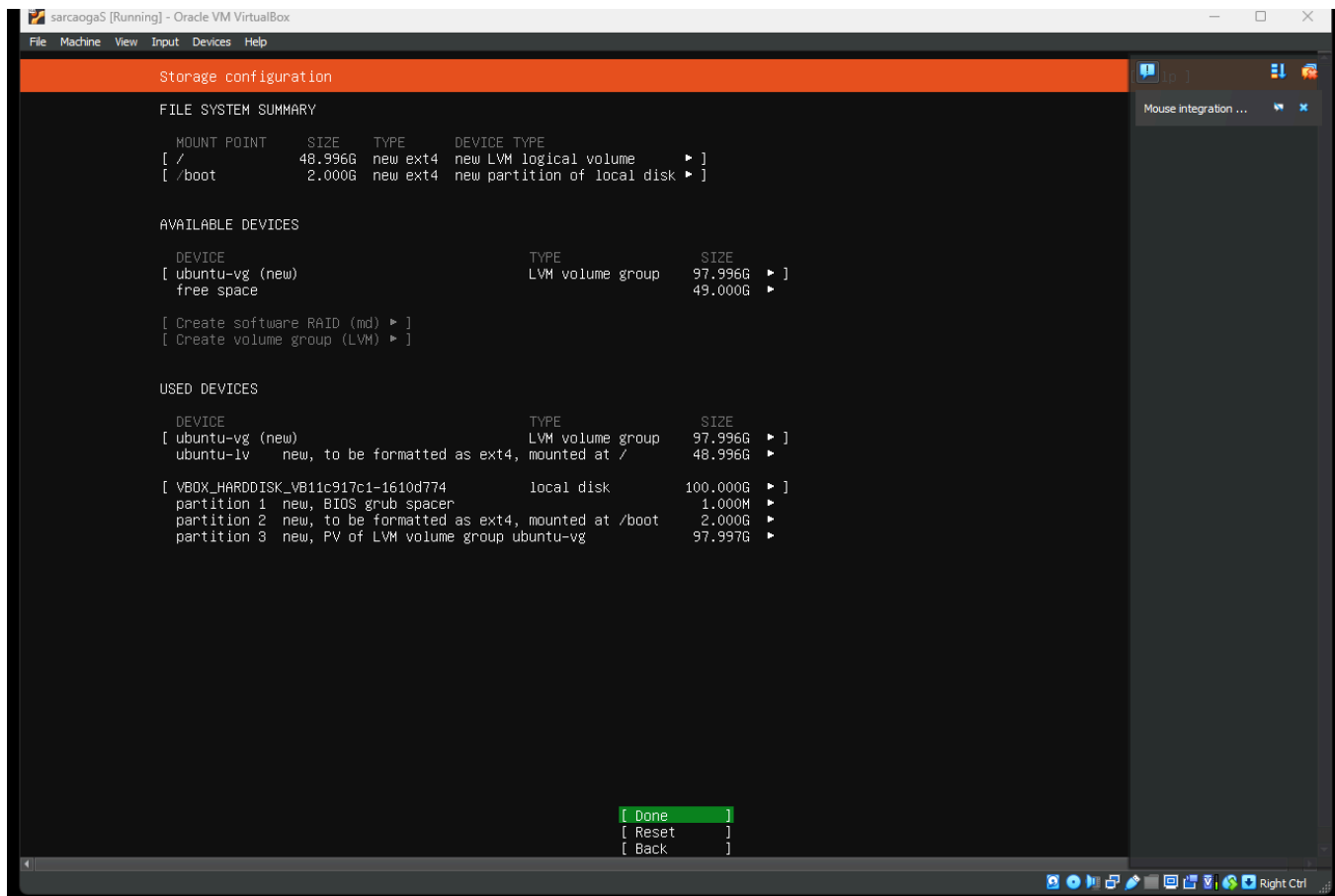


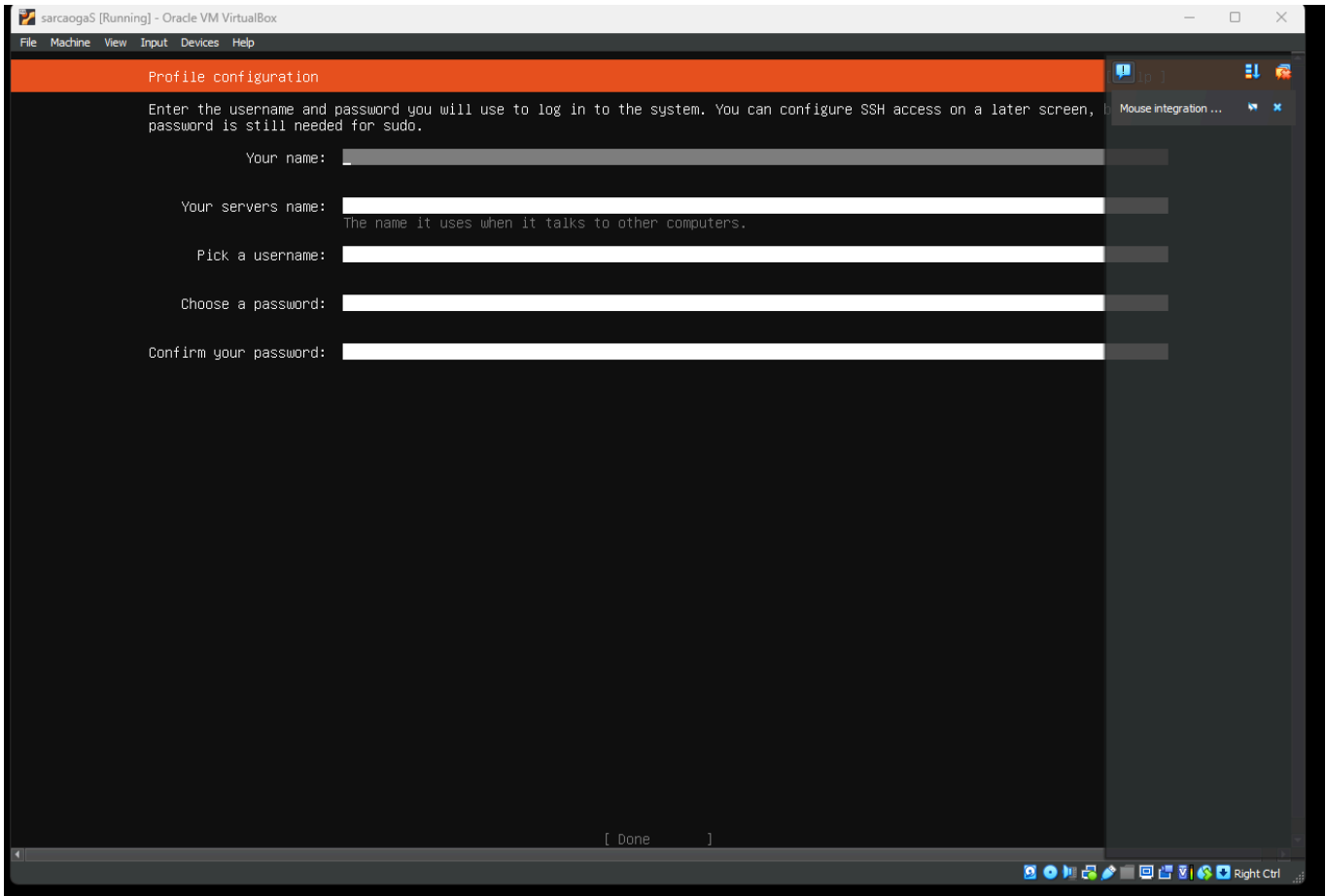


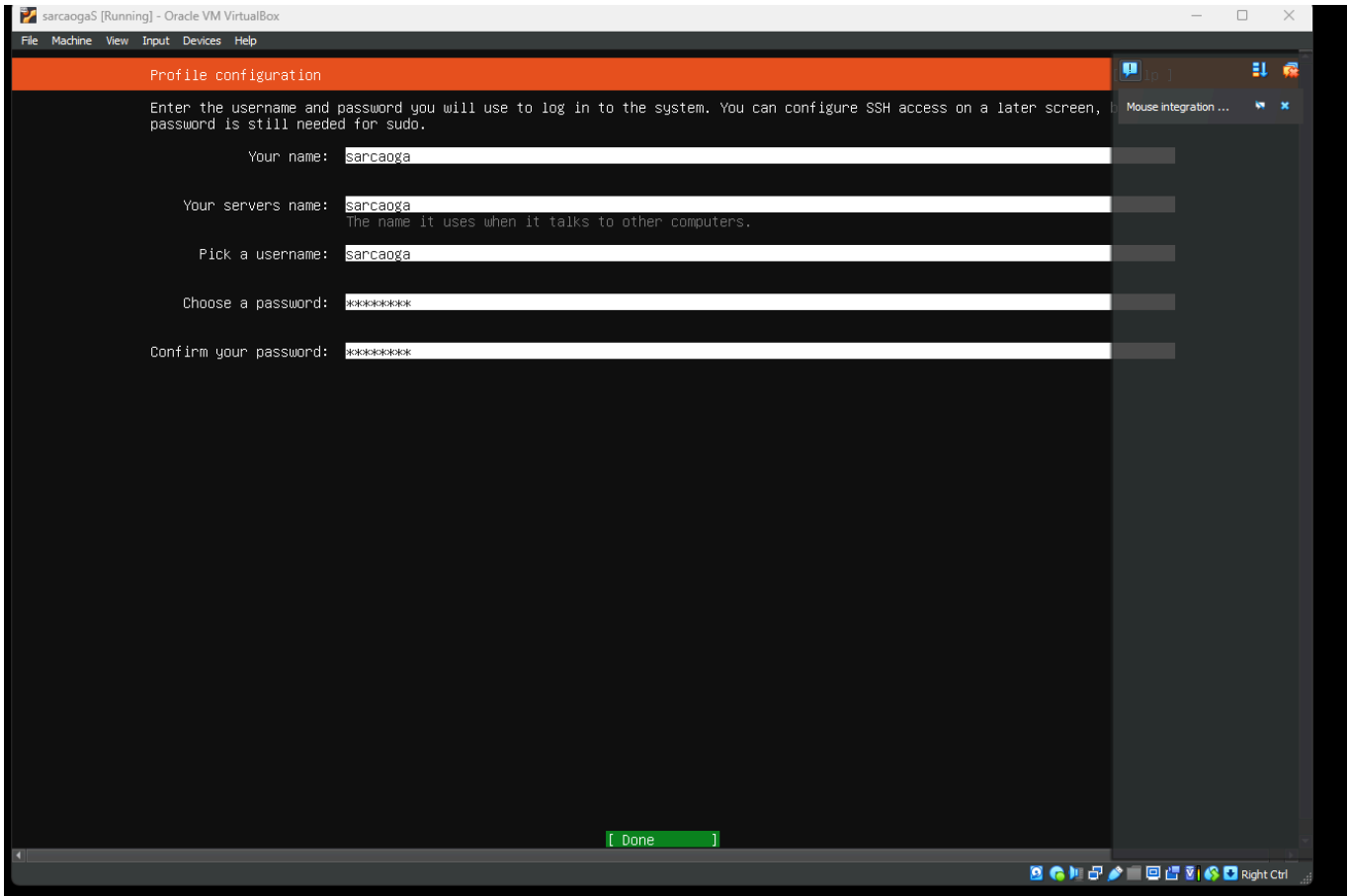


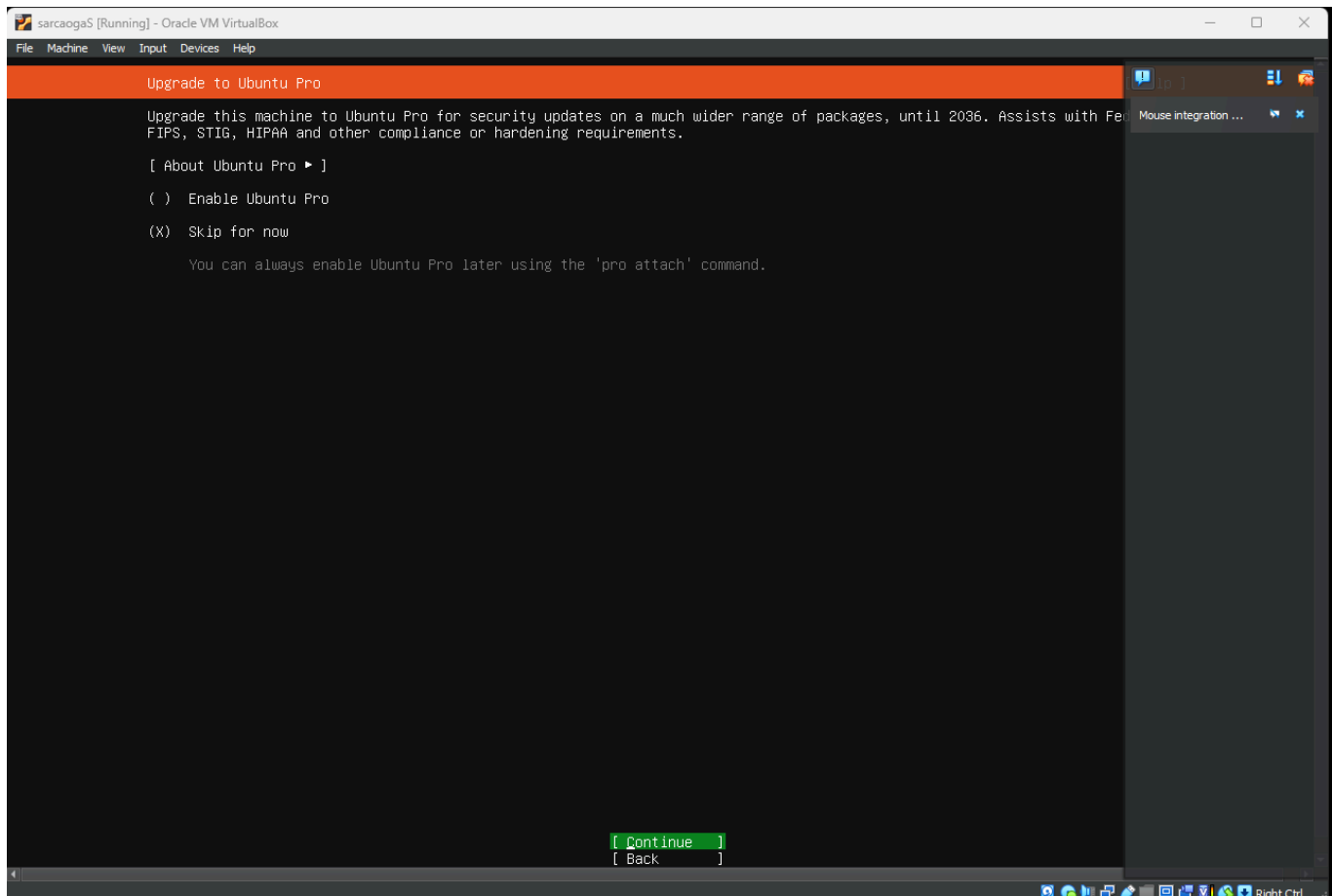


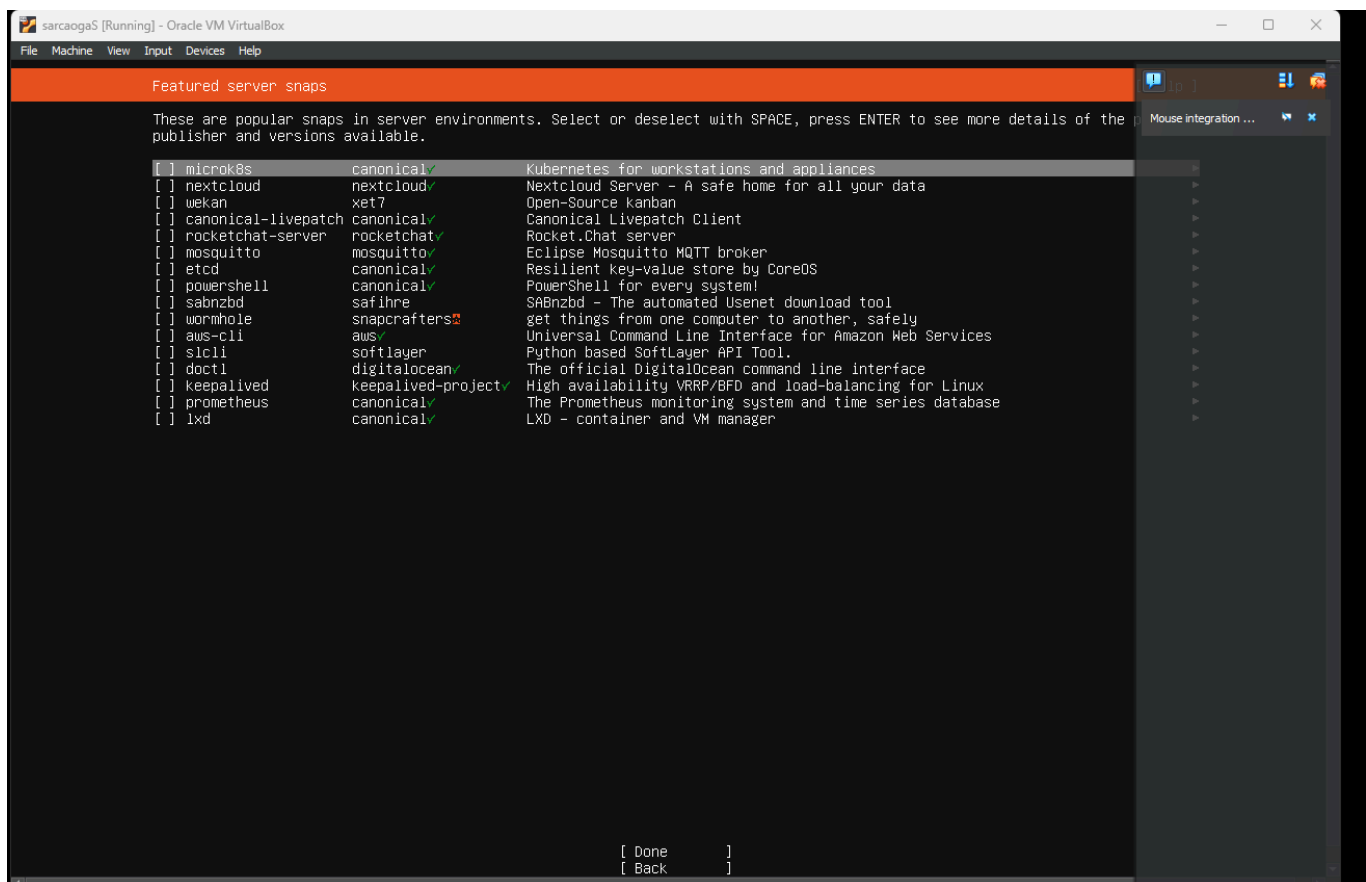
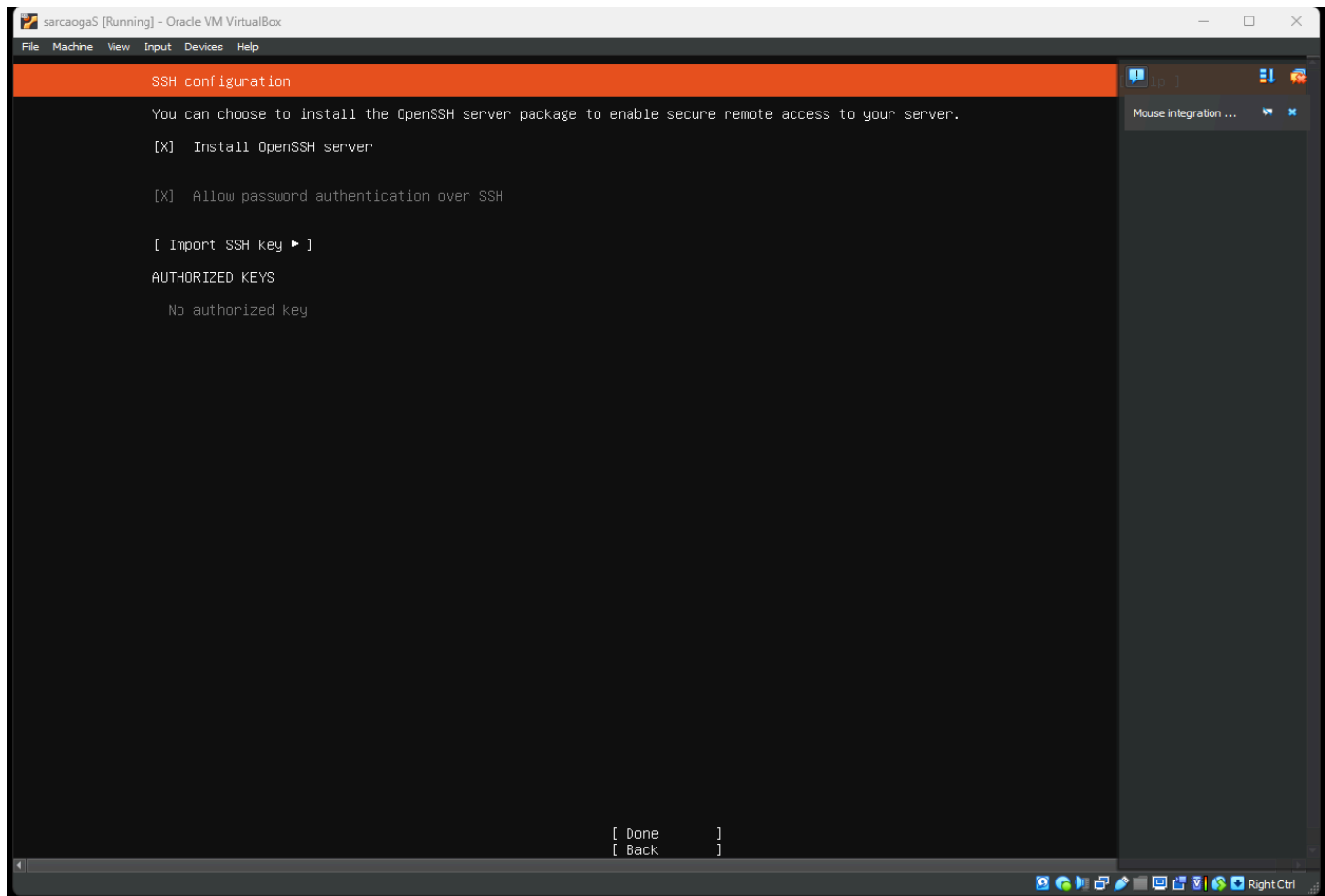


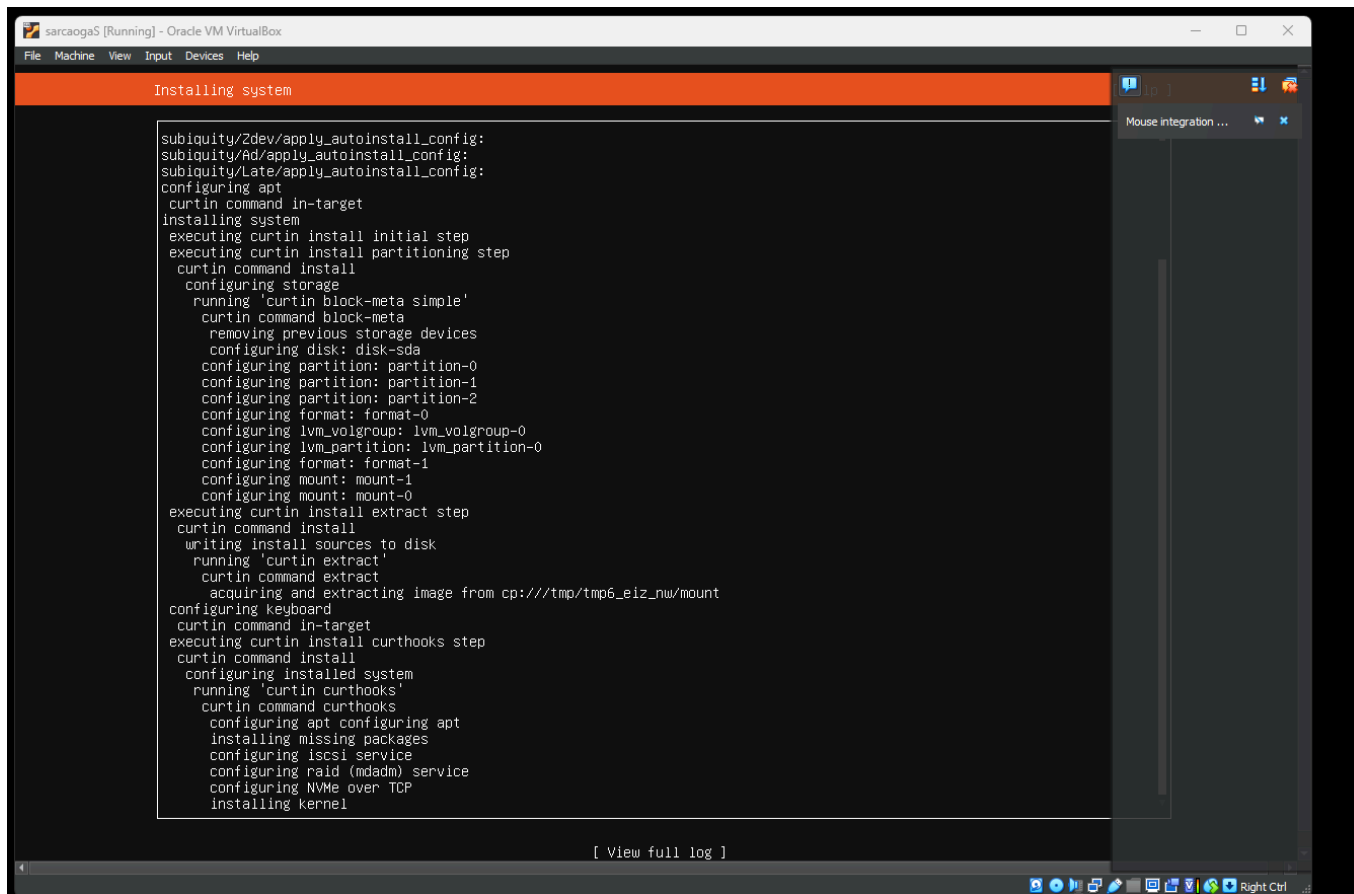


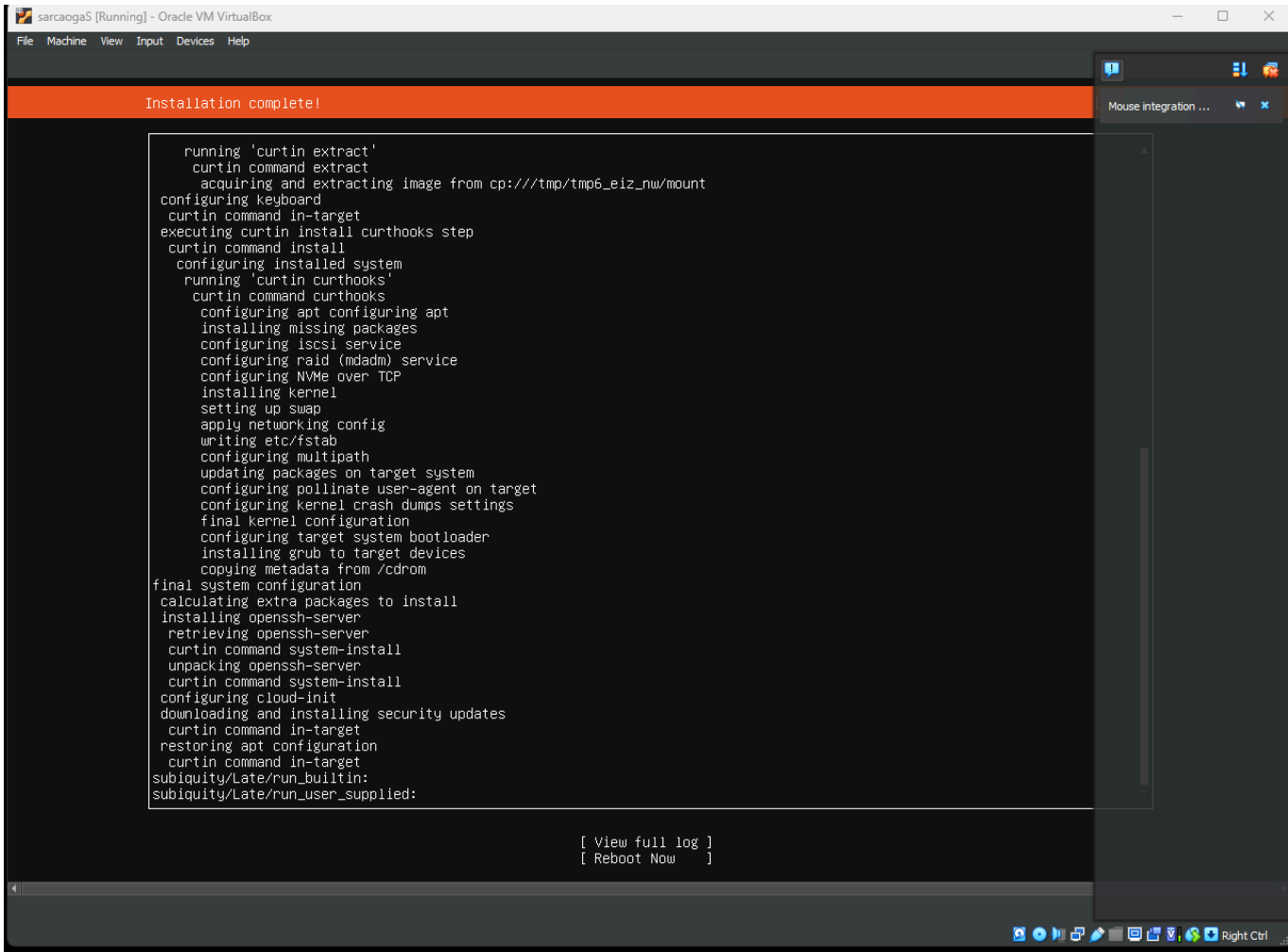












```
sarcao9a5 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Ubuntu 26.04 LTS sarcao9a tty1
sarcao9a login: sarcao9a
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Wed Sep  2 08:24:25 AM UTC 2026

System load:  0.13           Processes:            157
Usage of /:   15.0% of 47.93GB Users logged in:      0
Memory usage: 5%           IPv4 address for enp0s3: 192.168.137.89
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

sarcao9a@sarcao9a:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:89:39:03 brd ff:ff:ff:ff:ff:ff
    altname enx080027893903
    inet 192.168.137.89/24 metric 100 brd 192.168.137.255 scope global dynamic enp0s3
        valid_lft 604794sec preferred_lft 604794sec
    inet6 fe80::a00:27ff:fe89:3903/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
sarcao9a@sarcao9a:~$ _
```